

Orcasound Young Explorers

Help students investigate real underwater recordings and spectrograms, think like bioacousticians, meet J, K, and L pods, and choose one action that helps Southern Resident Killer Whales.

1. Hear

Investigate sound evidence

2. Meet

Explore three local pods

3. Protect

Choose an Orca Ally action

Designed for grades 3-8

Flexible 5-, 15-, and 30-minute formats. No student account, upload, public post, or personal information is required.

Prepare the room

What you need

- A computer with internet access
- Projector or student devices
- Speakers or headphones at a comfortable volume, or the visual-only spectrogram path
- This guide or the student observation sheet

Before students arrive

1. Open the Young Explorers HTML activity.
2. Test one audio clip and the volume.
3. Choose Ocean Explorer or Junior Bioacoustician.
4. Decide whether students work together or independently.

Learning objective

Students will use an audio recording and/or spectrogram to make one evidence-based observation about a Southern Resident call, explain one limit of that evidence, and choose one relevant conservation action.

Students will also be able to:

- describe a hydrophone as an underwater microphone;
- make an observation from an audio clip without overstating certainty;
- explain that sound can support pod recognition but usually does not identify one whale;
- distinguish an AI candidate from an expert-confirmed detection;
- connect quieter water, healthy Chinook salmon, and clean waterways to orca recovery.

Essential vocabulary

Term	Student-friendly meaning
Hydrophone	An underwater microphone that records sounds traveling through water.
Bioacoustics	The study of sounds made by living things and the places where they live.
Spectrogram	A picture showing how a sound's pitch and strength change over time.
Pod	A family-based social group of orcas.
Echolocation	Sending clicks and listening for echoes to find objects or prey.
Evidence	Information used to support an explanation.

Science guardrail: Loudness alone does not give an exact whale location. Calls can help experts recognize a pod, but identifying an individual usually requires a suitable dorsal-fin and saddle-patch photograph reviewed by a qualified source.

Population scope: This activity covers fish-eating Southern Residents. Bigg's killer whales also use the Salish Sea, primarily hunt marine mammals, and are often quieter while hunting.

Three ways to teach it

5-minute launch: Hear

1. **1 minute:** Ask, "How could we know an orca is nearby if we cannot see it?"
2. **3 minutes:** Complete one Ocean Explorer question as a class.
3. **1 minute:** Exit prompt: "What did you observe, and what are you not yet sure about?"

15-minute exploration: Hear + Meet

1. **2 minutes:** Introduce hydrophones and comfortable listening volume.
2. **7 minutes:** Complete the five-question quiz using audio, spectrograms, or both.
3. **4 minutes:** Compare the representative J/S01, K/S16, and L/S19 calls.
4. **2 minutes:** Collect pod stamps and share one evidence-based observation.

30-minute investigation: Hear + Meet + Protect

1. **3 minutes:** Hook and vocabulary.
2. **10 minutes:** Complete and discuss the quiz.
3. **7 minutes:** Compare all three pod calls using the observation sheet.
4. **7 minutes:** Teams investigate quiet water, healthy salmon, or clean water.
5. **3 minutes:** Each student chooses an Orca Ally action and completes an exit ticket.

Opening script

SAY: Today we are ocean sound detectives. Orcasound uses underwater microphones called hydrophones to listen to the Salish Sea. We can investigate recordings, real spectrograms, or both. We will use "not sure" when evidence is limited and discover how careful review can help people protect Southern Resident Killer Whales.

Discussion norms

I notice...

Describe what you actually hear or see.

I think...

Offer an explanation based on evidence.

I wonder...

Name what more you would need to know.

Hear, Meet, Protect

Mission 1: Hear

- Remind students that nothing autoplays and they control pause and volume.
- Invite every student to use the recording, matching spectrogram, or both; hearing is not required for success.
- Ask for observations before interpretations: patterned, tonal, repeated, steady, rising, falling, bright, or faint.
- Celebrate "not sure" as a responsible scientific response.
- After each reveal, ask: "What evidence supports that explanation?"

SAY: A recording is evidence from one listening location. It tells us what reached that hydrophone; it does not automatically tell us the exact location or identity of a whale.

Mission 2: Meet

Pod	Representative call	Listening prompt	Important reminder
J Pod	S01	Notice the short tonal pattern.	A call can support pod recognition.
K Pod	S16	Does it remind you of a mewling sound?	A familiar call is not an individual ID.
L Pod	S19	Compare its pitch and pattern.	Some call types may be shared.

These examples sit within Orcasound's larger learning ladder of 46 characterized Southern Resident call types, S01-S46. The prototype starts with S01-S05, then uses S01, S16, and S19 for pod comparison.

SAY: Sound and photographs answer different questions. Calls may help experts recognize a pod. Researchers usually identify an individual using dorsal-fin shape, nicks, scars, and saddle-patch patterns in suitable photographs.

Mission 3: Protect

Quiet water
Vessel noise can interfere with communication and echolocation used during hunting.

Healthy salmon
Southern Residents specialize in fish and strongly prefer Chinook salmon.

Clean water
Reducing pollution supports healthier habitat, prey, and marine food webs.

Student choices are learning commitments, not operational reports. Student answers must never create vessel alerts, public sightings, or confirmed detections.

Ocean Sound Detective Notes

Name or team (optional): _____ Date: _____

1. My sound observation

☐ patterned call ☐ flat tone ☐ changing pitch ☐ not sure

2. Observation versus inference

I directly noticed...

I think it might mean...

3. Compare two pod calls

Call	Pitch or tone	Pattern or rhythm	One question I have

4. My evidence confidence

☐ I need more evidence ☐ Some evidence supports this ☐ Strong evidence supports this

5. My Orca Ally action

☐ quiet water ☐ healthy salmon ☐ clean water

Ocean Explorer: grades 3-5

#	Question	Answer	Teaching point
1	What do you notice?	A short, patterned call	Discrete calls are short tonal patterns used in communication.
2	What does a hydrophone do?	Records sound underwater	A hydrophone records acoustic evidence at its location.
3	What can louder audio tell us?	The whale may be closer, but other factors matter	Orientation, depth, sound type, water, noise, and equipment also affect volume.
4	How do researchers identify one whale?	A clear dorsal-fin and saddle-patch photograph	Qualified researchers compare visible patterns with a verified catalog.
5	What happens after AI hears a possible call?	Ask a trained human to review the sound	Orcasound volunteer moderators review candidates; an AI candidate is not an expert-confirmed detection.

Suggested follow-up prompts

1. Why is "not sure" a useful answer for a scientist?
2. What can a hydrophone tell us? What can it not tell us by itself?
3. Why might the same whale sound louder or quieter at different times?
4. What is one difference between a sound clue and a photo clue?
5. Why should a person review an AI candidate?

Accept thoughtful language: Students do not need the exact scientific vocabulary if they can describe the observation accurately. Reward careful evidence use rather than confident guessing.

Quick scoring interpretation

Score	Teacher response
4-5	Invite the student to explain evidence or try Junior Bioacoustician.
2-3	Replay selected clips and compare observations with explanations.
0-1	Model one question together; emphasize listening comfort and "not sure."

Junior Bioacoustician: grades 6-8

#	Question	Answer	Teaching point
1	Classify the mystery sound	A discrete, tonal call	Classification should use the recording and spectrogram together.
2	What does a spectrogram add?	Frequency and intensity over time	A spectrogram visualizes sound structure, not exact location or identity.
3	Can one call prove which pod is present?	It can support identification with context	Shared calls and recording quality limit certainty.
4	What provides stronger location evidence?	Synchronized signals from multiple known sensors	Arrival-time differences can support localization when the system is suitable.
5	Which AI workflow is responsible?	Candidate -> expert review -> confirmed status -> approved use	Preserve provenance, uncertainty, and review status.

Extension: evidence ladder

Ask teams to rank these statements from weakest to strongest evidence, then revise each claim:

1. "The audio was loud, so the whale was exactly 100 meters away."
2. "A listener thought the sound resembled an orca call."
3. "An AI model flagged a possible call."
4. "A trained reviewer confirmed an orca call in the recording."
5. "Multiple synchronized sensors and reviewed sightings support a pod-level interpretation."

SAY: Good science keeps the observation, source, uncertainty, and review status visible. It does not turn every clue into certainty.

Extension: design challenge

Explain that students are practicing a simplified version of the evidence review done by Orcasound's volunteer moderators. Have students sketch a responsible detection card with:

- hydrophone name
- date and time
- audio clip
- spectrogram

- evidence label
- review status
- confidence or uncertainty
- source attribution

Make the activity work for everyone

Accessibility

- Treat real spectrogram observation and text descriptions as a complete investigation path, not a fallback.
- Keep volume comfortable and never require headphone use.
- Read prompts aloud when helpful.
- Allow replay, extra time, partner work, and "not sure."
- Do not rely on color alone to communicate an answer.

Privacy

- No student account is required.
- Do not ask students to publish names or locations.
- Certificate text lasts only for the current page session.
- Do not upload student work without school approval.
- Open external resources with an adult.

Live-listening expectations

Live hydrophones may be quiet or may capture vessels, weather, and other animals. Silence is not a failed lesson. Use curated historical clips for predictable instruction and live audio for optional observation.

Shared devices and kiosks: Use the visual path when a room is noisy, start over before the next group, and close or refresh the page so a typed certificate name is not left for the next visitor.

Trusted further study

- **Orcasound live listening:** <https://live.orcasound.net/>
- **Orcasound learning library:** <https://www.orcasound.net/learn/>
- **J, K, and L representative calls:** [Orcasound favorite calls lesson](#)
- **NOAA killer whale overview:** <https://www.fisheries.noaa.gov/species/killer-whale>
- **NOAA sound and hunting:** [Sound Is a Strategy for Hunting](#)
- **NOAA salmon and orcas:** [Salmon & Orcas teacher resources](#)
- **Be Whale Wise:** <https://www.bewhalewise.org/>

Content review note: This packet supports a concept prototype. Orcasound acoustic and education subject-matter experts should review scientific wording, clip labels, and classroom recommendations before production publication.